Graphing linear inequalities in two va[redacted]les

1

a No

b Yes

2

a

i No

ii No

iii No

iv Yes

b

i No

ii No

iii No

iv Yes

c

i No

ii No

iii No

iv Yes

3

a $y \leq 5$

b $x \geq 2$

c $y \geq -2x - 6$

d $y < 2x + 6$

e $y > -3x$

f $y \leq 2x + 5$

g $y \geq 3x - 3$

h $y < -2x - 4$

i $y \leq -2$

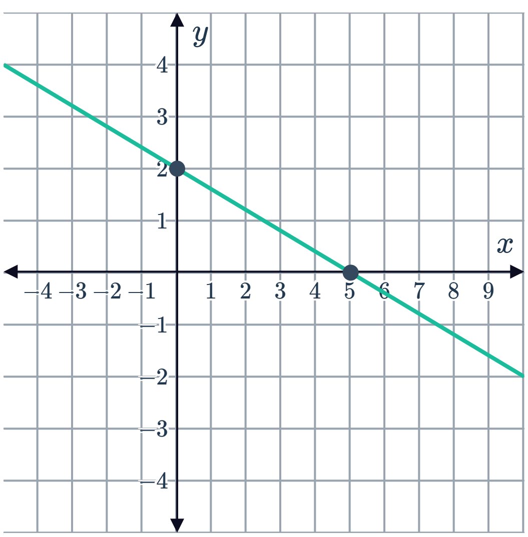
j $y > -3x + 6$

4

a $x = 5$

b $y = 2$

c



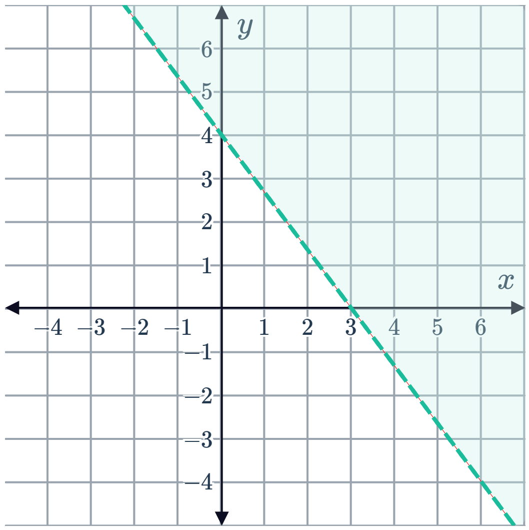
d $2x + 5y > 10$

5

a $x = 3$

b $y = 4$

c



d No

6

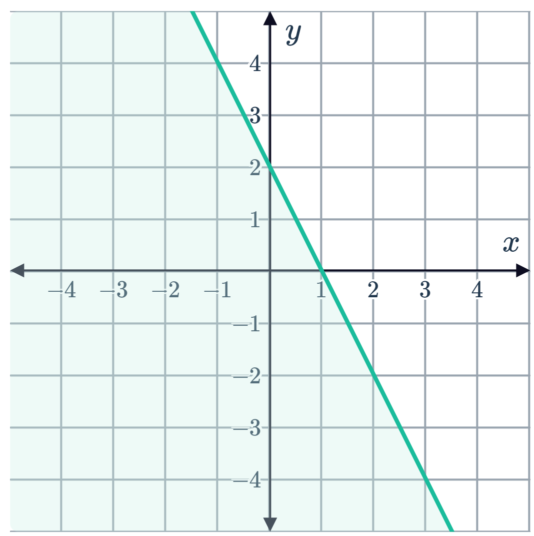
a $x = 1$

b $y = 2$

c

i No **ii** Yes **iii** No **iv** No

d



e Yes

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7

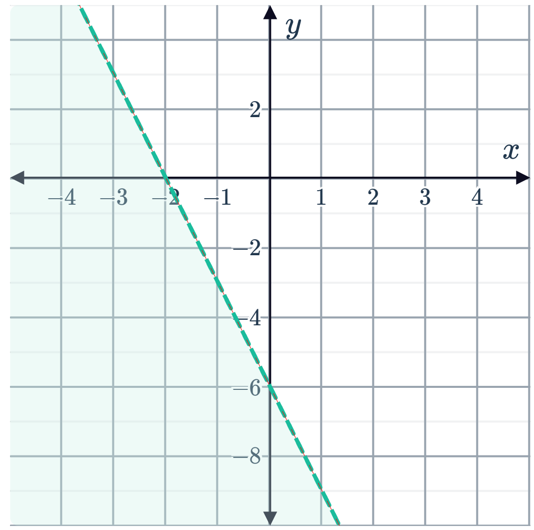
a $x = -2$

b $y = -6$

c

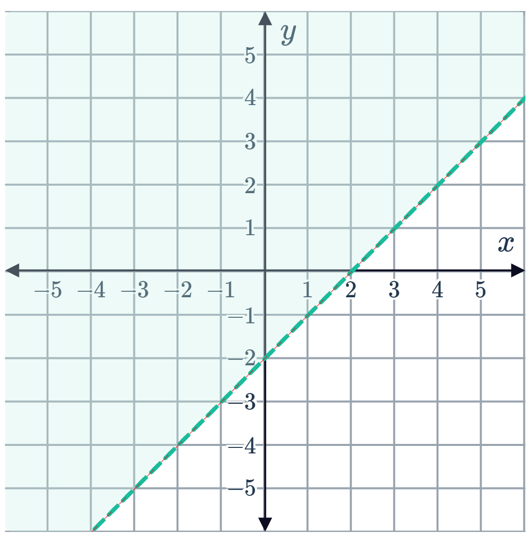
i No **ii** No **iii** Yes **iv** No

d

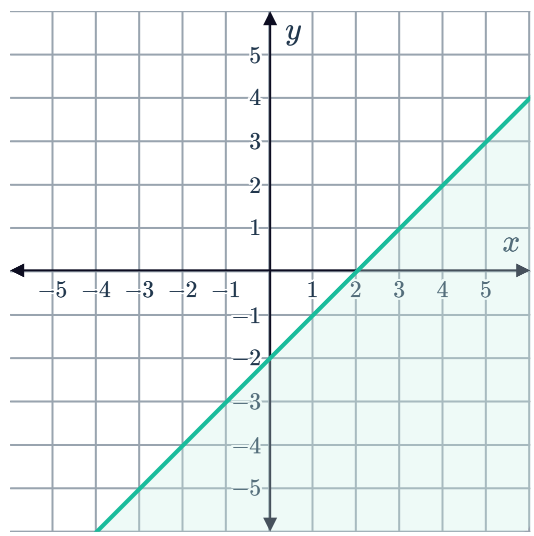


8

a



b



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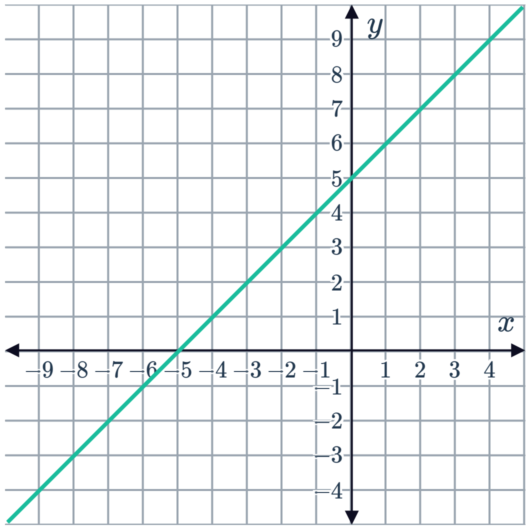
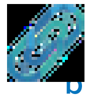
9

a



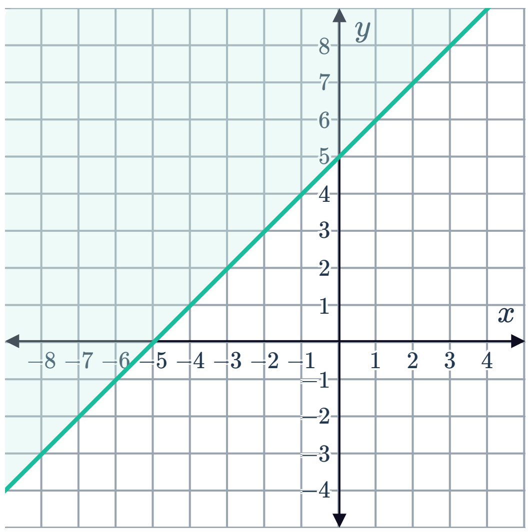


b

i Yes

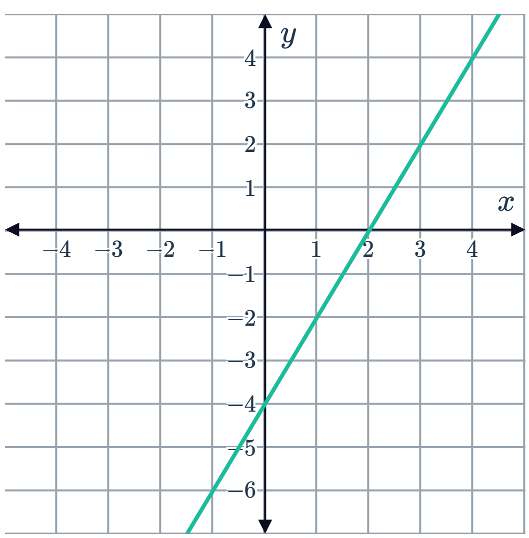
ii No

c



10

a



b

i No

ii Yes

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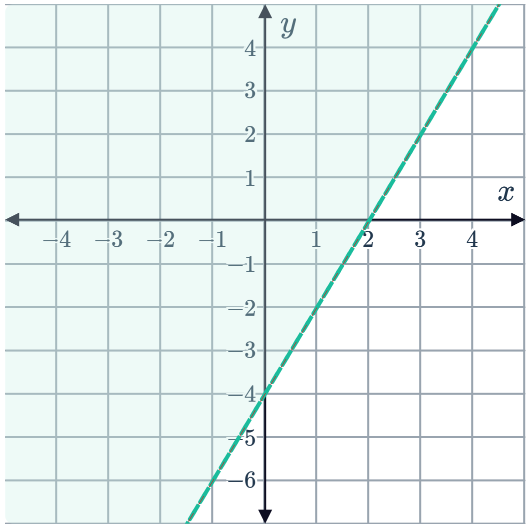
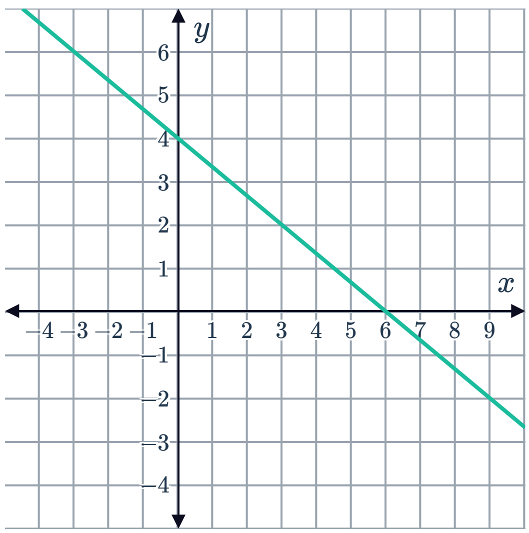
c



11

a

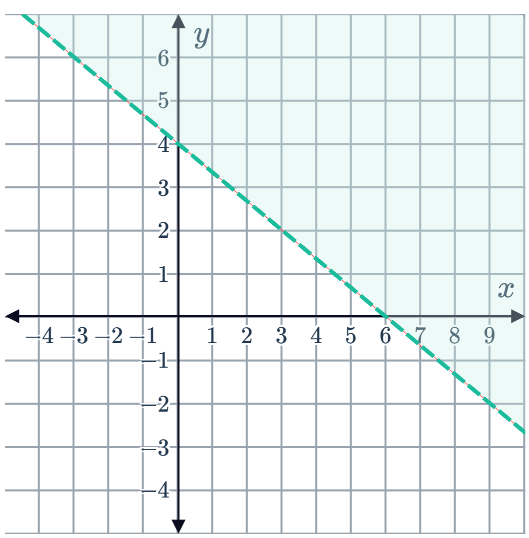


b

i No

ii Yes

c



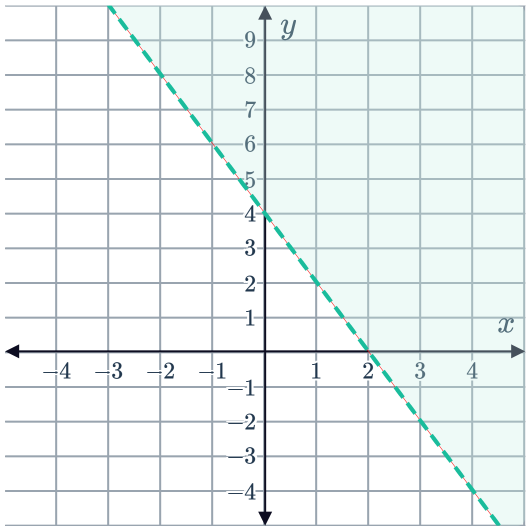
12

a

x	0	1	2
y	4	2	0

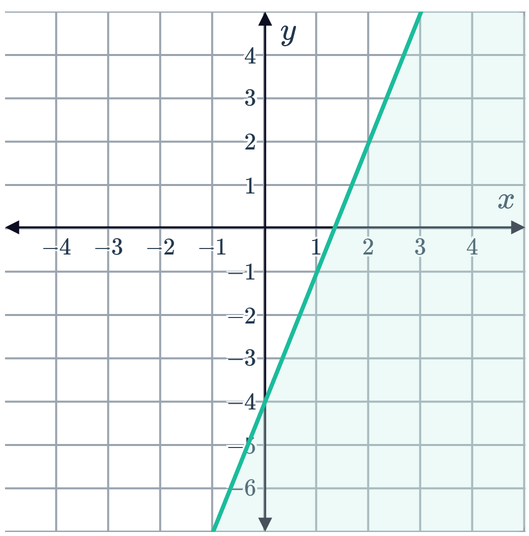
b **i** No **ii** Yes **iii** No **iv** No

c

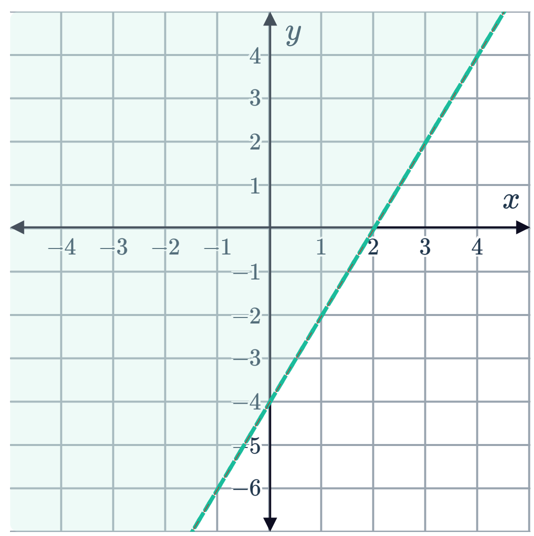


13

a



b



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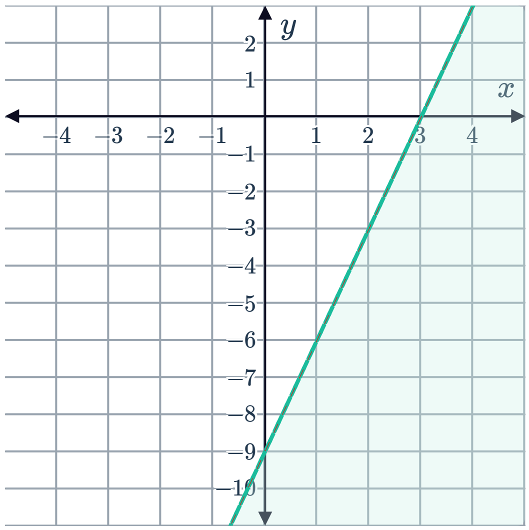
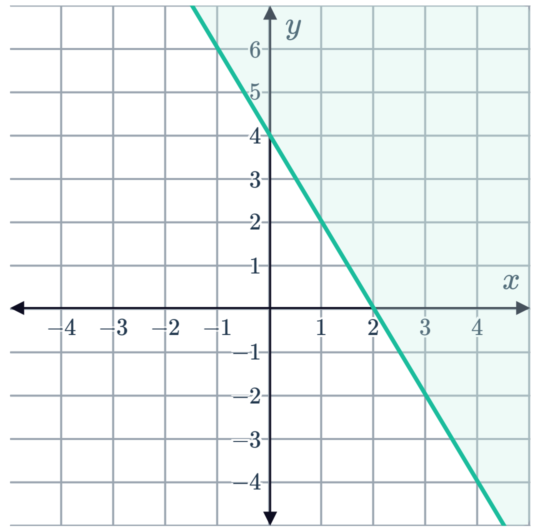
c





14

a

i 4 ii $y = 4x - 5$ iii $y \leq 4x - 5$

b

i 3 ii $y = 3x - 6$ iii $y \geq 3x - 6$

15 $y \leq 95 - \dfrac{19}{9}x$

16 $y \leq 75 - \dfrac{15}{17}x$

17 $y \leq x - 58$

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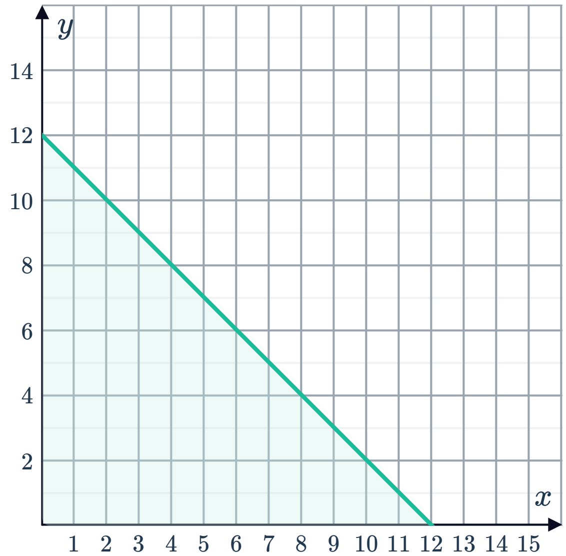
18

a $y \leq 12 - x$

b



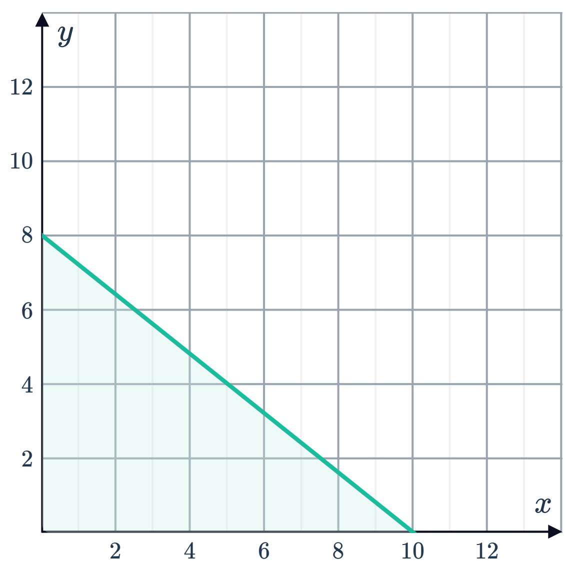
c

i No **ii** Yes **iii** No **iv** No

19

a $y \leq 8 - \frac{4}{5}x$

b



c

i No **ii** Yes **iii** No **iv** No

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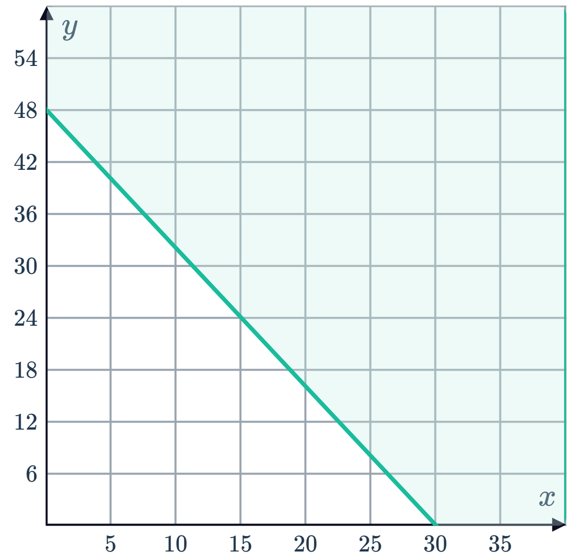
20

a $y \geq 48 - \frac{8}{5}x$

b



c

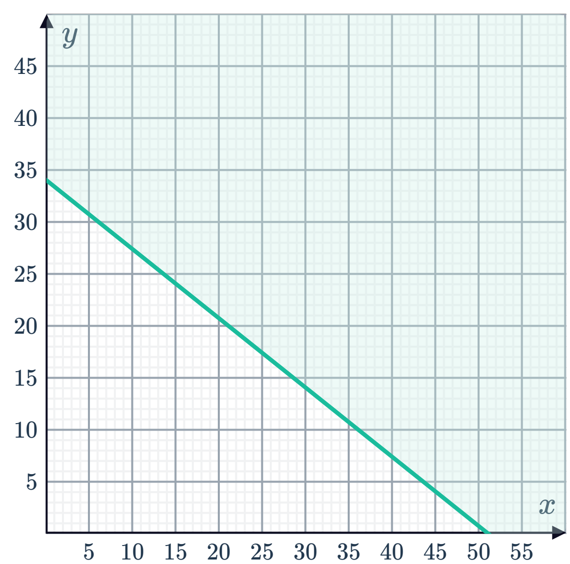
i No **ii** No **iii** No **iv** Yes

d 30 books

21

a $y > 34 - \frac{2}{3}x$

b



c **B**

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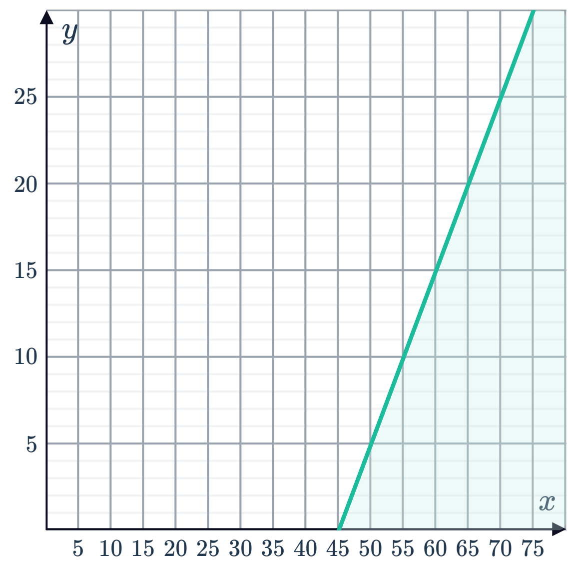
22

a $y \leq x - 45$

b



c 65 minutes

Additional questions

23

a Solid

b Dashed

c Solid

d Dashed

24

a $(2, 4)$

b $(2, 4)$, $(-3, 2)$ and $(-1, -6)$

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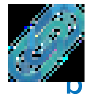
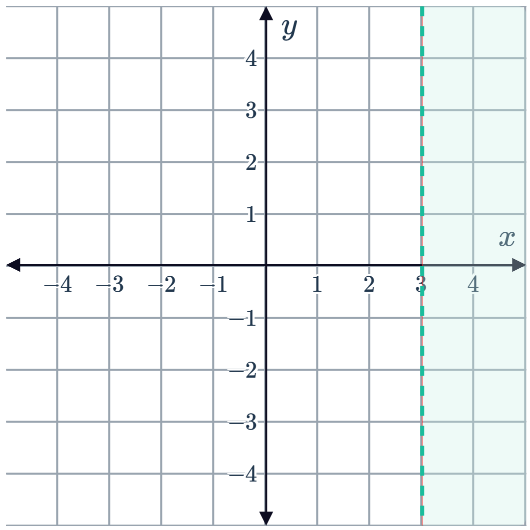
25



a

i The point $(0,0)$ does not satisfy the inequality.

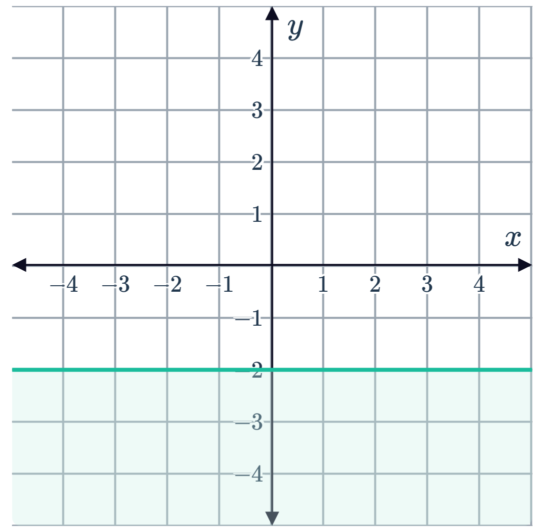
ii



b

i The point $(0,0)$ does not satisfy the inequality.

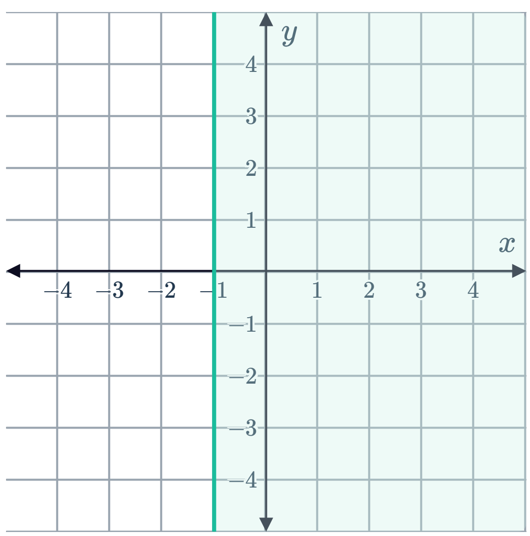
ii



c

i The point $(,0,0)$ satisfies the inequality.

ii



26 $x+y \geq 43$

27 $10x+13y \geq 260$

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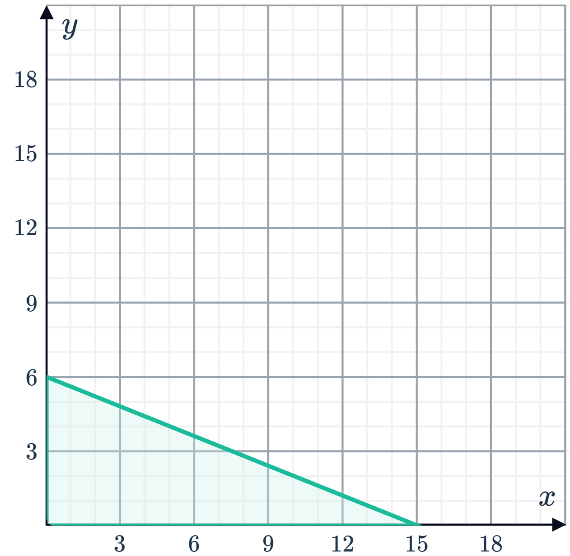
28

a $2x + 5y \leq 30$

b



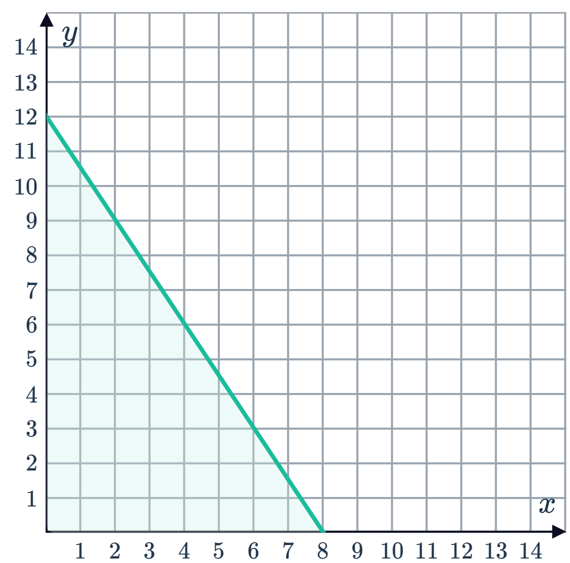
c The x-intercept of 15 represents the greatest number of games of Table Tennis that Oprah could play if she decides not to play any games of Frog Days.

d The y-intercept of 6 represents the greatest number of games of Frog Days that Oprah could play if she decides not to play any games of Table Tennis.

29

a $6x + 4y \leq 48$

b



c There are 61 different ways, one for each integer coordinate inside the solution set of the inequality.